

Raaghav Thirumaligai

Mechanical Engineer

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OBJECTIVE

Controls-focused mechanical engineer (PhD '30) with hands-on mechatronics and vehicle systems experience (FSAE EV). Built real-time control and estimation projects in MATLAB/Simulink (inverted pendulum: swing-up + LQR/LQG) and developed instrumentation + automation workflows (Arduino-MATLAB DAQ, Python + vision-based detection).

Seeking controls / automation internship roles in the Bay Area (model-based design, control theory, real-time).

SKILLS

Control Theory – State-space Linear Systems, Lyapunov stability, LQR/LQG, MIMO analysis
Nonlinear Control – Stability analysis via class K functions; feedback linearization, input-output ideas
Model Predictive Control – Constraint handling, finite-horizon optimization, trajectory optimization
Estimation / Filtering – Observers, Kalman filtering (LQG); sensor calibration, DAQ, filtering for experiments
Decision / Game Theory – Zero-sum / bimatrix games, saddle-point concepts; Minimax Theorem; dynamic games
Robust Control – System norms, small-gain theorem, H^∞ synthesis, mixed sensitivity
Hybrid Systems – Continuous-discrete modeling, hybrid analysis & design
Programming – MATLAB/Simulink (System ID and Control); Python (ML / sensors); Arduino; Git
CAD / Analysis – SolidWorks & Fusion 360; GD&T; assemblies; drawings; FEA
Fabrication – Rapid prototyping, fixtures, waterjet/laser, sheet metal, 3D printing (FDM/SLA)

EDUCATION

- **University of California, Santa Barbara - Ph.D. Mechanical Engineering** Projected Jun 2029
GPA 3.91/4.0 — Research Interest: Control Theory
- **University of California, Santa Barbara - M.S. Mechanical Engineering** Jul 2025
GPA 3.90/4.0 — Project: [Clogging of fiber suspensions at constrictions](#)
- **University of California, Santa Barbara - B.S. Mechanical Engineering** Jun 2025
GPA 3.76/4.0 — Senior Capstone: [600 V Vehicle Battery Container](#)

PROJECTS

Inverted pendulum on cart: swing-up + balancing + catch (MATLAB/Simulink) Apr 2025 - Jun 2025

- Identified linear models around stable and unstable equilibria using logarithmic sine sweeps and MATLAB System Identification (tfest); validated via frequency response/root-locus checks.
- Implemented a state-machine controller: energy-based swing-up, LQR/LQG (Kalman filter) balancing near upright, and LQR/LQG "catching" to return to the downward equilibrium.
- Demo + writeup: [video](#) | [project page](#)

GauchaRacing (Formula SAE EV): vehicle systems + accumulator container Aug 2021 - Jun 2025

- Led vehicle systems work as Internal VP / VP; 4-year contributor on electric race car design-build-test.
- Senior capstone: designed and built a 600 V accumulator container (adhesive-first joining strategy); produced CAD, drawings, and structural checks.
- Braking system lead: designed/manufactured components; supported testing and reliability improvements.
- Led an undergraduate research grant project to characterize brake-fluid viscosity (rheometer study); presented results at an undergraduate research colloquium.

Automated clog detection & reset testbed (Python control + vision monitoring) Jul 2024 - Oct 2025

- Prototyped a vision-based clog detector using a rolling 5 s brightness metric to identify flow occlusion events from video.
- Integrated detector output into Python test scripts for automated logging;

RESEARCH EXPERIENCE

Biological Control Lab, Yeung UCSB - *Research Assistant* Jan 2025 - Jan 2026

- Co-developed a control-oriented model of genome-wide transcription coupled to DNA supercoiling dynamics regulated by topoisomerases.
- Supported analysis of stochastic and nonlinear population models (stability/sensitivity analysis) to identify parameters influencing transcription and growth dynamics.
- Co-author on manuscript accepted (Jan 2026); preprint: [10.1101/2025.05.26.655193](#).

M&M Flow Lab, Sauret UCSB - *Master’s Project* Aug 2024 - Jun 2025

- Researching clogging dynamics in particle-laden suspensions in constricted flows, relevant to medical/industrial systems.
- Prototyped a vision-based clog detector using a rolling brightness/windowed-visibility metric; used it to support logging and post-analysis of clogging events.
- Presented findings at November 2024 APS Division of Fluid Dynamics meeting.

M&M Flow Lab, Sauret UCSB - *Undergraduate Researcher* Jan 2023 - Aug 2024

- Explored cohesive forces in granular materials with two PhD student mentors to help understand inter-particle mechanics from the introduction of cohesive forces.
- Designed and built an instrumented annular shear cell; developed an Arduino–MATLAB DAQ + calibration workflow for strain-gauge measurements and repeatable experiments.
- Presented work at Southern California Flow Physics Symposium.

Fluid Energy Science Lab, Luzzatto-Fegiz UCSB - *Undergraduate Researcher* Jun 2023 - Oct 2023

- 3D-printed and laser-cut annuli to simulate kites for wind energy applications.
- Data collection for wake model verification in a wind tunnel; made an Arduino-MATLAB DAQ system and app.

EXPERIENCE - WORK

Teaching Assistant - *UCSB: Mech E Academic Student Employee* Apr 2024 - Present

• Grading and holding office hours for Vibrations course covering signals and dynamics	Jan - Mar 2026
• Led Mechanical Engineering Laboratory Section covering experimental procedures and data analysis	Sept - Dec 2025
• Taught Dynamics Office Hours covering FBD and Energy based system analysis	Apr - Jun 2025
• Graded assignments and advised bio-inspired design groups	Jan - Mar 2025
• Assisted in grading for Dynamics course and Fluids course	Apr - Dec 2024

Resident Assistant - *UCSB: San Rafael Residence Hall* Aug 2022 - Jun 2023

- Managed resident support, conflict resolution, and event planning in a high-traffic dorm environment.

GameMaster - *Diablo Escapes, Walnut Creek* Jul 2020 - Aug 2022

- Guided customers through VR escape rooms; troubleshoot hardware/software issues and explained unfamiliar tech.
- Helped design puzzles for physical escape rooms under construction.

HONORS

• Undergraduate Engineering Honors Program	Sept 2023 - Jun 2024
• Dean’s Honors (ENGR) x5 per quarter	
• Undergraduate Research Awards x2 (\$3750 and \$2500)	2024 and 2025